

Another Good Reason to Become a Vegetarian

Our focus in *Home Energy* has always been on energy-efficient, healthy, affordable, and sustainable homes, but there are other areas of our lives that account for our total carbon footprint—how much carbon we are responsible for adding to the atmosphere—a measure of our contribution to global warming. Our houses and apartments, but also our cars, air travel, and the food we eat, all contribute.

Don Fugler, who does research for the Canada Housing and Mortgage Corporation, estimated the amount each area of our lives contributes to our carbon footprint. He used a hypothetical family of four (two adults, two kids) in Ottawa, with a medium-sized house (2,400 square feet) and two cars (Ford Explorer, 20 mpg highway and 16 mpg city; and Honda Fit, 40 mpg highway and 30 mpg city) to do the calculations. A mid-efficiency gas furnace and a natural draft gas water heater heat the home—the family spends about \$1,200 per year on heating air and water. Both parents work, and each one travels about 20 miles round trip to work each weekday. The kids travel a few miles each day back and forth to school. The two parents make a total of five trips to Toronto and five trips to other places each year for business. The family flies round trip to Whistler once a year to go skiing and travel round trip by car to visit relatives in Nova Scotia once a year.

Our hypothetical family, according to Fugler's calculations, emit about 11.5 tonnes of CO₂ from their house for heating, hot water, appliances, and lighting; about 13 tonnes from the fossil fuel used in air travel; about 9 tonnes from their cars; and about 4.5 tonnes from the food they eat (including growing, shipping, and waste disposal).



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That's about 38 tonnes total per year. (A tonne, or metric ton, is 1,000 kilograms, or 2,205 lb. A ton, or short ton, is 2,000 lb.) Notice that the highest amount is from air travel! Fugler did not include embodied energy in these calculations; the embodied energy of the house and cars is a relatively small portion of the family's total carbon footprint.

How would technological changes affect our hypothetical family's carbon footprint? If the family installed renewable energy systems to meet all of their electricity needs, or if their local utility started using 100% hydropower, that would reduce their carbon emissions by a little over 3 tonnes, or about 7%. If, instead of installing renewable energy sources, the family did an energy retrofit of the home, including a new high-efficiency furnace and water heater, new refrigerator and freezer, lighting retrofit with CFLs and LED lights, and some air sealing and added insulation (costing approximately

\$8,000), they would shave 4.5 tonnes off their carbon footprint, bringing it down from 38 tonnes to about 33.5 tonnes.

Meanwhile, the family would save about 40% on their household utility costs.

But what about lifestyle changes! If Mom and Dad took the bus to work every weekday, they could shave off another 3 tonnes. If the family moved to an efficient town house downtown, skipped the ski trip, and cut some business travel, they would lose another 10 or so tonnes. If they really made some serious changes by going to a zero energy downtown house, walking everywhere, and becoming vegetarians, they could reduce their carbon footprint by 11 more tonnes, leaving a total footprint of only 10 tonnes of carbon.

If you are a regular *Home Energy* reader, you have probably already made your home much more energy efficient than the average home. But what about making some lifestyle changes? Fugler recommends that we conduct more and more of our business using the Internet, instead of traveling far from home; live close to our jobs in dense urban areas with good public transportation; ride our bikes a lot; and buy more local food and less meat.

—Jim Gunshinan

Jim Gunshinan is Home Energy's managing editor.

For more information:

The folks who brought us the movie *An Inconvenient Truth* also provide an online calculator so that you can calculate your contribution to global warming. The site also gives good information on how to reduce your carbon footprint. Go to www.climatecrisis.net/takeaction/carboncalculator.

For more details on Don Fugler's calculations, contact him at dfugler@cmhc-schl.gc.ca.